Literature-driven extraction and computational prediction of causal statements linking genetic variants to biological processes, pathways and phenotypes

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Introduction

- ❖ Pathogenic variants influence disease through cascading molecular mechanisms that bridge genotype to clinical phenotype
- ❖ Literature-derived causal networks capture the biological context essential for understanding variant-disease relationships
- ❖ Integration of mechanistic pathway knowledge with computational tools enables detailed and more accurate variant interpretation

Methodology

Extract causal pathways from the literature

INDRA Database

"... **A53T α-synuclein** overexpression increased LDH release (296%, p < 0.001), **caspase 1** (157%, p = 0.001) and caspase 3 (157%, p = 0.017) activities..."

"**Caspase 1** can be activated by known inflammasome-stimulators to directly cleave α-synuclein to generate truncated species that are prone to form aggregates in a neuronal **PD** cell model, and silencing of **caspase 1** expression rescues neurotoxicity caused by α-synuclein, suggesting that under certain toxic stimuli, inflammasome, and **caspase 1** cause **PD** pathology in neurons…"

[PMID: 33494411]

Causal Path

SNCA A53T

↓ [Activation]

CASP1

↓ [Activation]

Parkinson's Disease

Variant feature construction

M P S T E D V

Sequence: MDVFMK…EPEA (140 AA)

Embedding by ESM2 (x_{ESM2})

A → T

Variant AA (x_{AA}): A→T

Variant Position(x_{pos}): 53

Domain feature(x_{dom}): REGION

ClinVar

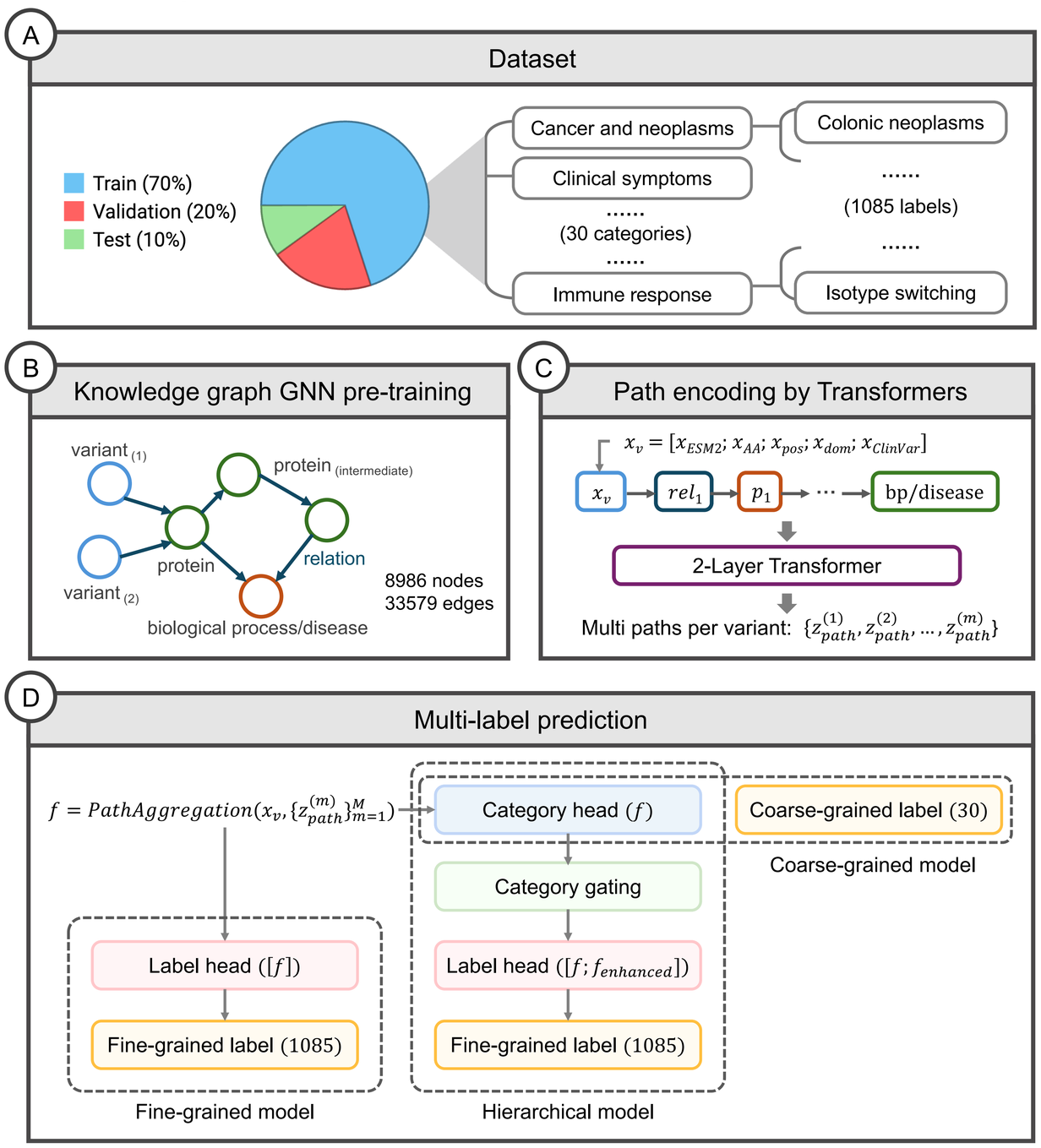
Pathogenicity: Pathogenic

Review: ★ (single submitter)

Embedding for ($x_{ClinVar}$)

SNCA A53T (x_v)

Hierarchical prediction framework



Result

Protein variant network

- **25,862** variant-to-disease pathways across **2,561** proteins with interactive visualization
- Evidence tracking for each causal edge
- Interactive portal: http://variants.indra.bio

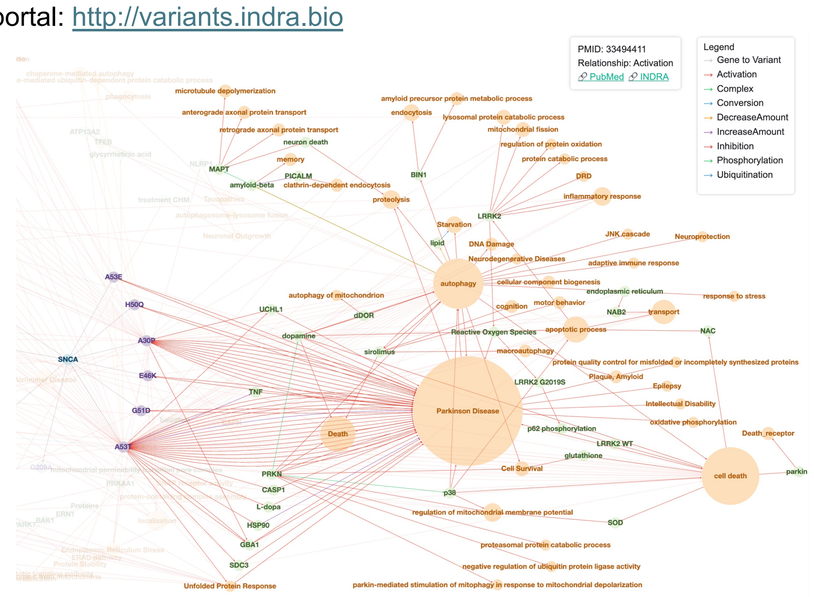


Fig 1. Interactive visualization of SNCA variant network

Hierarchical model improves prediction performance

- Categories represent 30 broad functional classes
- Labels represent 1085 specific diseases/biological processes
- The hierarchical approach leverages category predictions to enhance fine-grained label classification

Table 1. Average test performance across 30 data splits for the three prediction strategies

Predictor	Micro-F1	Micro-P	Micro-R	AUPRC	AUROC
Coarse-grained model	0.908	0.959	0.863	0.751	0.970
Fine-grained model	**0.647**	**0.684**	0.622	0.286	0.913
Hierarchical model	0.573	0.520	**0.644**	**0.307**	**0.919**

De novo variant impact prediction: biological and disease processes

Variant	VPS35_R524W		PSEN1_L435F	
Query	Sequence	MPTTQQSP...PIYEGLIL (796 AA)	Sequence	MTELPAPLSYF... QLAFHQFYI (467 AA)
	Domain	REGION	Domain	REGION, MOTIF
	ClinVar	Pathogenicity: not provided Review: 0	ClinVar	Pathogenicity: not provided Review: 0
Predict	biological process/disease	prob	biological process/disease	prob
	translation (protein synthesis degradation)	62.9%	localization (cellular transport)	74.1%
	DNA damage (DNA chromatin processes)	49.7%	translation (protein synthesis degradation)	62.5%
	Parkinson disease (neurological disorders)	49.0%	apoptotic process (cell death survival)	51.0%
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Discussion

- ❖ Combined literature pathways with hierarchical machine learning for mechanistic variant interpretation and accurate prediction
- ❖ Integrated protein sequences, ClinVar annotations, and domain features to predict variant-disease associations for novel variants
- ❖ Portal (https://variants.indra.bio) and model provide mechanistic insights and computational tools for variant analysis

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